FMSE-DGAF: Frequency-Modulated Spectral Enhancement and Dynamic Global–Local Attention Fusion Network for Hyperspectral Image Classification

Kun Hu, Xiaoqing Wan, *Member, IEEE*, and Hui Liu

***Abstract*—Hyperspectral image (HSI) classification is a challenging task owing to high dimensionality, spectral redundancy, limited labeled samples, and the presence of mixed pixels. Although models based on convolutional neural networks (CNNs), Transformers, and hybrid architectures have improved feature extraction capabilities, they remain limited in simultaneously capturing complementary frequency information, fine-grained local details, and long-range global context. To address these challenges, this article proposes the frequency-modulated spectral enhancement and dynamic global–local attention fusion (FMSE-DGAF) method, which integrates three key modules: a frequency-modulated spectral enhancement module (FMSEM), a global–local feature fusion module (GLFFM), and a dynamic weighted convolutional attention module (DWCAM). First, the FMSEM strengthens discriminative spectral structures by modulating frequency-domain amplitudes while preserving phase information. Second, the GLFFM integrates global contextual information with fine-grained local spatial structures, stabilizing intermediate spatial representations and effectively aligning coarse-grained semantic cues with fine-grained spatial representations. Finally, the DWCAM combines local convolutional enhancement with global self-attention to achieve multiscale and multigranularity spectral–spatial feature fusion, while emphasizing informative features and suppressing redundancies. Comprehensive evaluations on five benchmark HSI datasets with a wide spatial resolution range (0.109–30 m, approximately a 300-fold variation) show that FMSE-DGAF delivers consistently competitive classification performance, attaining overall accuracies of 99.89%, 99.96%, 99.91%, 99.89% and 99.76% on the Pavia University, Salinas, WHU-Hi-LongKou, WHU-Hi-HanChuan, and Botswana datasets, respectively. These results verify the effectiveness, robustness, and strong generalization capability of the proposed method for HSI classification tasks.**

***Index Terms*—Classification, dynamic global–local attention fusion, frequency-domain enhancement, hyperspectral image (HSI).**

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I. Introduction

HYPERSPECTRAL image (HSI) has become an indispensable technology in modern remote sensing due to its ability to simultaneously capture rich spectral signatures and detailed spatial information across hundreds of contiguous bands [1]. By integrating spectral signatures with spatial context, hyperspectral imagery enables reliable characterization of diverse materials and land-cover types, which underpins numerous applications such as environmental monitoring [2], precision agriculture [3], mineral exploration [4], and urban analysis [5]. However, HSI classification remains a challenging task because of several intrinsic factors, including high dimensionality, spectral redundancy, limited labeled samples, and the presence of mixed pixels [6]. These challenges often cause performance degradation, particularly in complex or highly heterogeneous scenes.

Early research in HSI classification primarily relied on traditional machine learning models such as support vector machines (SVMs) [7], [8], K-nearest neighbors (KNNs) [9], [10], [11], and random forests (RF) [12], [13]. These methods demonstrated strong performance when sufficient spectral separability existed, and dimensionality reduction techniques like principal component analysis (PCA) or linear discriminant analysis (LDA) were applied [14], [15]. However, traditional models often treat each pixel independently, failing to exploit spatial context. Meanwhile, handcrafted features are insufficient to model the intricate spectral–spatial relationships and nonlinear distributions commonly found in hyperspectral data. As a result, their classification accuracy and generalization ability become limited in realistic scenarios.

With the development of deep learning [16], [17], [18], convolutional neural networks (CNNs) have become one of the most widely used methods for HSI classification [19], [20]. CNNs effectively extract spatial textures and local structures through convolutional kernels. Various CNN-based architectures, including 1-D, 2-D, and 3-D CNNs, have been proposed to model spectral, spatial, and joint spectral–spatial information [21], [22]. While these methods achieve notable improvements, they are inherently constrained by the locality of convolution operations, making it difficult to capture

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long-range dependencies and global contextual information. Increasing network depth or kernel size can partially address this issue, but often leads to higher computational cost and reduced efficiency, particularly under limited training samples [23].

To address the limitations of convolution, Transformers have recently gained attention in HSI classification [24], [25]. Transformers leverage self-attention mechanisms to model global relationships across spectral bands and spatial regions, achieving remarkable improvements in long-range dependency modeling and feature interaction. However, these methods, despite their strong capability in global modeling, may incur increased computational cost due to the use of self-attention mechanisms and often exhibit limited effectiveness in capturing fine-grained local structures [26].

In summary, despite the remarkable progress of CNN-based, Transformer-based, and hybrid-based methods in HSI classification, existing methods still present certain limitations. Conventional convolutional approaches are effective at capturing local spatial structures but struggle to model long-range dependencies, whereas attention mechanisms excel at global modeling while being less sensitive to fine-grained local details. Hybrid methods attempt to alleviate these issues by combining convolution and attention; however, most existing approaches still operate predominantly in the spatial domain and lack explicit mechanisms to enhance discriminative spectral representations before high-level attention modeling. Beyond architectural limitations, hyperspectral data itself introduces additional challenges. Due to the inherent spectral variability and semantic ambiguity in real-world scenarios, different land-cover classes may exhibit highly similar spectral responses, while samples from the same class may present diverse spectral characteristics under varying imaging conditions. It should be noted that the mixed-pixel problem discussed in this work differs from the classical hyperspectral unmixing task. Traditional unmixing methods mainly focus on decomposing mixed pixels into endmembers and abundance fractions, whereas the problem considered in this work refers to classification ambiguity caused by spectral similarity, spatial overlap, and boundary interference among neighboring categories. From a representation learning perspective, such spectral overlap and boundary interference tend to produce ambiguous and less discriminative spectral–spatial representations, making it difficult for spatial-domain models to consistently capture discriminative patterns. Moreover, spatial-domain feature extraction may produce overly smooth feature representations in overlapping regions, thereby weakening local discriminative variations and reducing class separability. Frequency-domain analysis provides a complementary representation by transforming spectral–spatial information into the frequency space, where subtle structural variations and local boundary differences that may be weakened in the spatial domain can be reflected by distinct frequency responses. In particular, frequency-domain modulation can adaptively reweight different frequency components, enhancing informative spectral–spatial responses while reducing the influence of background interference and redundant variations. As a result, the learned representations become more discriminative, thereby improving spectral separability and alleviating classification ambiguity in spatially overlapping regions. Motivated by these observations, we propose frequency-modulated spectral enhancement and dynamic global–local attention fusion (FMSE-DGAF), a novel framework that integrates frequency-modulated spectral enhancement with dynamic global–local attention to progressively refine spectral–spatial representations. The following contributions are made in this work.

1) A frequency-modulated spectral enhancement module (FMSEM) is proposed to enhance spectral–spatial representations in the frequency domain. By modulating amplitude components while preserving phase information, which encodes structural and spatial arrangements, preserves geometric consistency, and is relatively invariant to intensity variations, thereby maintaining the intrinsic spatial–spectral structure of hyperspectral data, FMSEM strengthens discriminative spectral structures and improves robustness against spectral redundancy and noise in hyperspectral data.
2) A global–local feature fusion module (GLFFM) is designed to stabilize intermediate spatial representations by jointly modeling global contextual information and local structural details. Through dynamic convolution-based attention, GLFFM effectively aligns coarse semantic cues with fine-grained spatial representations.
3) A dynamic weighted convolutional attention module (DWCAM) is introduced to achieve unified local–global semantic modeling. By dynamically fusing convolutional inductive biases with multihead self-attention via learnable weights, DWCAM enables multiscale and multigranularity spectral–spatial feature aggregation.
4) Experiments are performed on five benchmark hyperspectral datasets: Pavia University, Salinas, WHU-Hi-Longkou, WHU-Hi-HanChuan, and Botswana. The experimental results indicate that the proposed FMSE-DGAF achieves competitive classification accuracy compared with several state-of-the-art methods, demonstrating its effectiveness and robustness.

The rest of this article is organized as follows. Section II presents a comprehensive review of related work. Section III details the implementation of the proposed FMSE-DGAF. Section IV reports the experimental results and parameter analysis conducted on five HSI datasets. Finally, Section V presents the conclusion.

II. Related Work

Driven by the rapid advancements in DL, CNNs, Transformer-based models, and, more recently, state-space architectures have significantly advanced HSI classification. In this section, we review these representative paradigms and discuss their strengths and limitations.

Early studies employed 1-D CNNs to extract interband correlations along the spectral dimension, modeling continuous spectral features at a relatively low computational cost [21]. Subsequently, 2-D CNNs were proposed to capture spatial textures, but they cannot simultaneously model spectral–spatial relationships [22]. 3-D CNNs, on the other hand, leverage

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3-D convolutional kernels to process spectral–spatial cubes, leading to notable improvements in classification accuracy; however, the resulting computational complexity and heavy reliance on abundant training data substantially constrain their practical scalability [23]. Building on this, recent research has explored more efficient and expressive hybrid convolutional structures to overcome the limitations of single convolutional paradigms [27], [28]. For instance, Ma et al. [29] proposed the ultralightweight hybrid CNN based on redundancy removal (ULite-R2HCN), a redundancy-removed lightweight hybrid convolutional network, which achieves ultralightweight design by eliminating redundant spectral and spatial convolutions and maintains effective spectral–spatial feature extraction under very limited sample conditions. To further balance complexity and expressive power, Ghaderizadeh et al. [30] proposed an integrated network design that couples a fast 3-D learning component with a 2-D convolutional backbone, enabling effective joint modeling of spectral and spatial characteristics while maintaining low computational complexity. In addition, Chen et al. [31] introduced the multichannel hybrid 2-D-3-D CNN (MH-2D-3D-CNN), which integrates 2-D and 3-D convolutions with spectral enhancement and attention mechanisms, significantly improving classification accuracy and stability under extremely limited labeled samples. Meanwhile, Shi et al. [32] proposed the hybrid CNN–GNN network (HCGN), which deeply integrates CNNs with graph convolutional networks (GCNs) and introduces an edge-enhanced module to model long-range structures and edge relationships, demonstrating superior classification performance in small-sample scenarios. Despite their success, the emphasis on local feature extraction in CNNs makes it difficult to adequately model interactions over extended spatial ranges. Increasing network depth or kernel size partially alleviates this issue but comes at the cost of higher computation and parameter complexity, making CNNs less efficient and more prone to overfitting under limited training samples [33].

Recently, Transformer-based models have gained attention in HSI classification due to their ability to model long-range dependencies and their flexibility in handling spectral–spatial data. For example, Zhang et al. [34] proposed the tensor Transformer (TT) framework, which employs tensor self-attention to directly process raw hyperspectral data, efficiently capturing long-range spectral–spatial features and avoiding the accuracy degradation caused by data serialization that disrupts structural information. Chen et al. [35] introduced the multigranularity spatial–spectral convolution Transformer (MSSCFormer) network, which integrates intraclass spectral attention, a spatial–spectral feature extractor, and a global–local convolutional Transformer to achieve granularity spectral–spatial feature extraction, reducing spectral variations among samples of the same class and fully exploiting both local and global information. Zhao et al. [36] developed the groupwise separable convolutional vision Transformer (GSC-ViT), a compact ViT framework that integrates groupwise separable convolutions with multihead self-attention to simultaneously capture local and global spectral and spatial cues, while controlling the number of parameters to improve classification performance under limited training data. Jia et al. [37] proposed the center mask Transformer (CMT), which incorporates multiscale convolutions and local–global embeddings along with regularized center mask pretraining, enabling efficient spectral–spatial feature extraction and further reducing model parameters, significantly improving classification accuracy under few-shot conditions. Zhang et al. [38] proposed TFTnet to address interdomain discrepancies in hyperspectral single-source domain classification, leveraging integrated spatial–spectral and frequency-domain features to achieve superior cross-scene performance.

Beyond CNNs and Transformers, state-space-based architectures, such as Mamba models, have recently attracted attention due to their ability to efficiently model long-range dependencies with linear computational complexity. He et al. [39] proposed 3DSS-Mamba, which captures global spectral–spatial features through selective scanning mechanisms, while Wang et al. [40] introduced S^2Mamba to enhance spatial–spectral feature fusion via structured state-space modeling. These methods demonstrate competitive performance in HSI classification and have been included as representative advanced baselines in this work. However, effectively modeling the intricate spectral–spatial dependencies and achieving adaptive feature representation across diverse HSI scenarios remain challenging.

III. Methodology

This section describes the FMSE-DGAF framework and its architectural design. The network comprises four components: the CNNs for initial feature extraction, the FMSEM for multiscale spectral–spatial feature enhancement, the GLFFM for global–local feature fusion, and the DWCAM for dynamic attention-based feature integration, as illustrated in Fig. 1.

A. Framework of FMSE-DGAF

FMSE-DGAF progressively extracts and refines spectral–spatial features using frequency-domain spectral enhancement and dynamic global–local attention, enabling effective discrimination in HSI. As illustrated in Fig. 1, the framework comprises three key components: an FMSEM, a GLFFM, and a DWCAM.

To begin with, PCA is applied to the raw HSI cube to reduce spectral redundancy and lower computational cost. The resulting compact representation is then fed into a 3-D CNN for initial extraction of combined spectral and spatial features. After 3-D convolutional feature extraction, the features are reshaped and refined using a 2-D convolutional layer, focusing on spatial feature modeling while maintaining the spectral characteristics captured earlier.

Although attention mechanisms are effective at capturing long-range spatial dependencies, their performance is heavily influenced by the quality of the underlying spectral features. Redundant or noisy spectral components at early stages can propagate through the network, potentially degrading class discrimination. Motivated by this observation, FMSE-DGAF introduces the FMSEM as the framework's core innovation. Operating in the frequency domain, the FMSEM adaptively modulates amplitude responses while preserving

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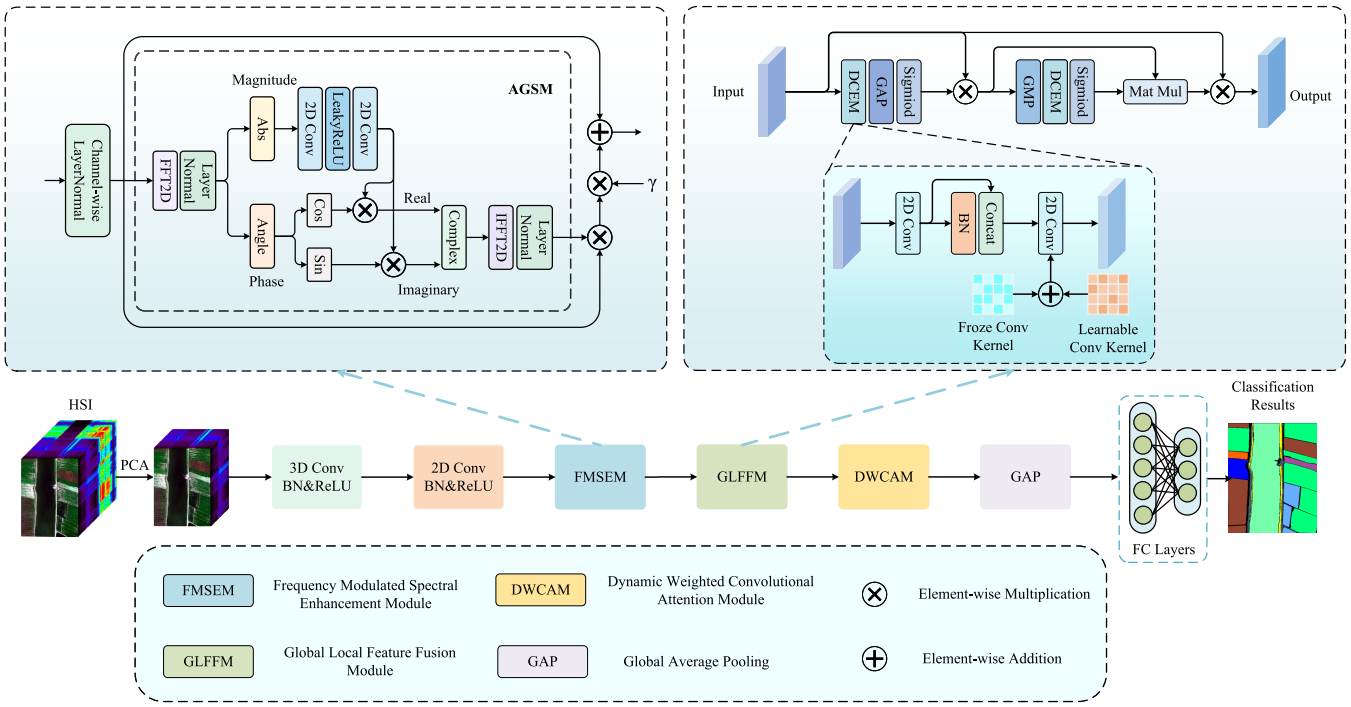


Fig. 1. Overall architecture of the proposed FMSE-DGAF, including PCA preprocessing for dimensionality reduction, 3-D/2-D CNNs for feature extraction, FMSEM for frequency-conditioned spectral enhancement, GLFFM for guided global-to-local fusion, and DWCAM for adaptive global–local attention integration.

phase information, thereby strengthening discriminative spectral structures before subsequent attention operations. As a result, subsequent spatial and semantic modeling is conducted on spectrally enhanced and reliable feature representations.

Following the FMSEM, the GLFFM stabilizes intermediate spatial representations by combining global contextual attention with local spatial sensitivity. Global attention captures channel-wise importance, while local attention highlights salient spatial regions. Their multiplicative fusion with the original features yields spatially coherent and semantically informative representations. These features are then fed into the DWCAM, which performs high-level semantic aggregation and long-range dependency modeling. The DWCAM dynamically integrates convolutional inductive biases with multihead attention through learnable fusion weights, ensuring that global semantic modeling operates on well-calibrated, spectrally enhanced, and spatially stabilized inputs.

Finally, the architecture incorporates an adaptive global average pooling operation, which is followed by a dropout stage and fully connected layers to perform pixel-wise classification. The dropout layer improves generalization by mitigating overfitting, while the fully connected layers transform the fused and attention-enhanced features into class-specific predictions.

The integration of the FMSEM, the GLFFM, and the DWCAM establishes a complementary and cooperative framework, with each module contributing distinct functionality: the FMSEM provides foundational spectral enhancement, the GLFFM stabilizes spatial representations and aligns multiscale contextual information, and the DWCAM performs high-level semantic aggregation. This structured collaboration allows the network to effectively address both spectral and spatial complexities inherent in HSI classification.

B. Structure of the FMSEM

The FMSEM is designed to enhance spectral–spatial feature representations in HSIs. It introduces a channel-wise layer normalization and an amplitude-guided spectral modulation (AGSM) mechanism in the Fourier domain, refining the magnitude components through adaptive modulation while preserving phase information (see Fig. 1). This allows the network to extract more stable and discriminative frequency cues. Additionally, an adaptive residual re-scaling strategy is applied to effectively integrate the frequency-refined responses with spatial-domain features. This design enables FMSEM to overcome the limitations of conventional spatial-only or spectral-only operators, improving robustness and HSI classification performance.

Specifically, given an input feature tensor $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, where B, C, H, and W denote the batch size, number of channels, height, and width of the feature map, respectively. The module first performs channel-wise normalization to reduce interchannel statistical variations. For each spatial location (h, w) and sample b, the mean $\mu_{b,h,w}$ and variance $\sigma^2_{b,h,w}$ along the channel dimension are computed as follows:

$$\mu_{b,h,w} = \frac{1}{C}\sum_{c=1}^{C} X_{b,c,h,w} \tag{1}$$

$$\sigma^2_{b,h,w} = \frac{1}{C}\sum_{c=1}^{C} \left(X_{b,c,h,w} - \mu_{b,h,w}\right)^2 \tag{2}$$

where $b \in \{1, 2, \ldots, B\}$ denotes the sample index, $c \in \{1, 2, \ldots, C\}$ denotes the channel index, $h \in \{1, 2, \ldots, H\}$

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denotes the height index, and $w \in \{1, 2, \ldots, W\}$ denotes the width index.

The normalized feature is then obtained through standardization and a learnable affine transformation

$$\hat{X}_{b,c,h,w} = \frac{X_{b,c,h,w} - \mu_{b,h,w}}{\sqrt{\sigma^2_{b,h,w} + \epsilon}} \cdot \gamma_c + \beta_c \tag{3}$$

where γ_c and β_c denote the channel-wise scaling and bias parameters, respectively, and ϵ is a small positive constant introduced for numerical stability.

To exploit frequency-domain information, the normalized feature $\hat{\mathbf{X}}$, denoted as $\mathbf{Z}$ for clarity in the AGSM, is first transformed along the spatial dimensions using a 2-D real-valued fast Fourier transform (rFFT2), yielding the frequency-domain representation

$$F = F(\mathbf{Z}) = \mathrm{rFFT2}(\mathbf{Z}). \tag{4}$$

For any batch b, channel c, and frequency coordinates (u, v), where (u, v) denote the horizontal and vertical frequency coordinates in the Fourier domain, respectively. The frequency coefficient can be expressed as follows:

$$F_{b,c}(u, v) = \mathrm{Re}_{b,c}(u, v) + j\,\mathrm{Im}_{b,c}(u, v) \tag{5}$$

where $j = \sqrt{-1}$ denotes the imaginary unit.

The frequency feature is then decomposed into amplitude and phase components. The amplitude is obtained from the real and imaginary parts as follows:

$$A_{b,c}(u, v) = \sqrt{\mathrm{Re}^2_{b,c}(u, v) + \mathrm{Im}^2_{b,c}(u, v)}. \tag{6}$$

The phase is calculated using the arctangent function

$$\Phi_{b,c}(u, v) = \arctan\left(\frac{\mathrm{Im}_{b,c}(u, v)}{\mathrm{Re}_{b,c}(u, v)}\right). \tag{7}$$

To achieve learnable amplitude modulation in the frequency domain, the amplitude A is transformed by a lightweight spectral MLP

$$A' = P(A). \tag{8}$$

It is worth noting that the proposed AGSM does not rely on any predefined frequency partition (e.g., hard separation of high- and low-frequency components). Instead, the modulation function $P(\cdot)$ operates on the full amplitude spectrum in a data-driven manner, enabling dynamic and continuous reweighting of different frequency components. Specifically, the learnable mapping adaptively adjusts the responses at each frequency location (u, v) conditioned on the input feature, enabling informative frequency patterns to be emphasized while reducing the influence of less informative responses. Benefiting from the global receptive property of the Fourier representation, the modulation process can effectively exploit structural and spectral variations distributed across different spatial locations, thereby facilitating the discrimination of confusing classes in overlapping regions. Moreover, AGSM reconstructs the spatial-domain feature using the modulated amplitude together with the preserved phase information, facilitating the preservation and enhancement of informative spectral–spatial structures while maintaining structural consistency and spatial details. This contributes to improved spectral separability and reduced classification ambiguity in regions with spectral overlap and boundary interference.

The mapping function is given by the following equation:

$$P(\cdot) = \mathrm{Conv}_{1\times1}(2C \to C)\left(\sigma\left(\mathrm{Conv}_{1\times1}(C \to 2C)(\cdot)\right)\right) \tag{9}$$

where $\sigma(\cdot)$ denotes the LeakyReLU activation function.

The modulated frequency feature is then reconstructed by combining the modulated amplitude with the original phase

$$\widetilde{F}_{b,c}(u, v) = A'_{b,c}(u, v) \cdot \left(\cos\Phi_{b,c}(u, v) + j \sin\Phi_{b,c}(u, v)\right). \tag{10}$$

Compared with amplitude information, which mainly reflects the energy distribution of frequency components, phase information preserves the spatial arrangement and structural organization of image features. In the frequency-domain representation, phase primarily encodes the positional relationships and geometric structures of spatial patterns, whereas amplitude mainly reflects the response strength of different frequency components. Therefore, preserving the original phase information during frequency-domain feature modulation helps maintain spatial consistency, edge continuity, and structural integrity, thereby mitigating spatial structural distortions during feature reconstruction and promoting structural consistency between the reconstructed features and the original spectral–spatial representations.

Finally, the inverse Fourier transform maps the modulated frequency feature back to the spatial domain, completing the frequency modeling and spatial reconstruction in the AGSM module

$$\widetilde{\mathbf{Z}} = F^{-1}\left(\widetilde{F}\right) \tag{11}$$

where $F^{-1}(\cdot)$ denotes the 2-D inverse real-valued fast Fourier transform (irFFT2).

The modulated feature $\widetilde{\mathbf{Z}}$ is then applied to the original input $\mathbf{X}$ via element-wise multiplication

$$\mathbf{X}' = \mathbf{X} \odot \widetilde{\mathbf{Z}}. \tag{12}$$

To retain the original representation while adaptively enhancing frequency-guided information, a learnable channel-wise scaling factor $\gamma \in \mathbb{R}^{1\times C\times1\times1}$ is introduced, and a residual connection is applied, yielding the final output

$$\mathbf{Y} = \mathbf{X} + \gamma \odot \mathbf{X}'. \tag{13}$$

To intuitively demonstrate the effectiveness of the proposed FMSEM, feature maps on the Pavia University dataset are visualized before and after applying the module, as shown in Fig. 2. The "before FMSEM" map exhibits relatively scattered feature responses with target features not well concentrated and significant background activation. In contrast, the "after FMSEM" map shows a much sparser distribution, where irrelevant noise is effectively suppressed, indicated by the darker regions. The AGSM mechanism effectively highlights discriminative spectral–spatial structures, amplifying response intensities at critical locations and making target features more prominent. This selective enhancement is further confirmed by the "Difference" map, where positive values align precisely with structural details, while negative values indicate the suppression of redundant background signals and irrelevant noise, demonstrating that the module adaptively reweights

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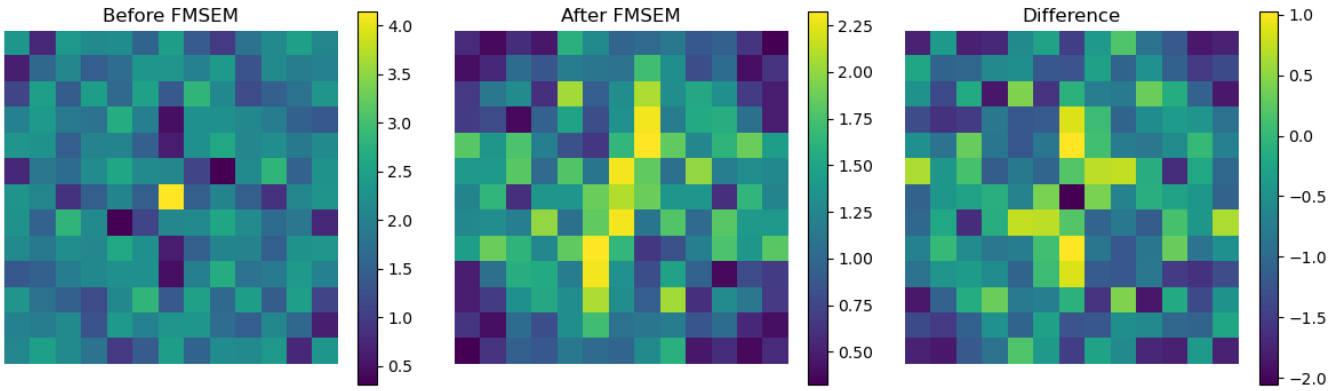


Fig. 2. Comparative visualization of feature maps on the Pavia University dataset. (Left) Original feature maps before FMSEM processing. (Middle) Feature maps after applying FMSEM. (Right) Difference map (after − before). Color intensity represents activation magnitude (dark blue: low, yellow: high).

informative frequencies rather than applying a uniform scaling. It is worth noting that to better reveal the internal distribution details of each stage, the color scales in Fig. 2 are normalized independently for each subfigure; this approach allows for a clearer observation of the relative activation patterns within each specific feature map, thereby highlighting the structural refinement achieved by FMSEM.

C. Structure of the GLFFM

The GLFFM is proposed to stabilize intermediate spatial representations by jointly modeling global context and local structural details (see Fig. 1). Using a dynamic convolution enhancement mechanism (DCEM), it adaptively adjusts convolutional kernels to emphasize informative features. A dual-branch attention structure captures information at multiple granularity levels. The global branch aggregates contextual cues, while the local branch focuses on position-sensitive fine-grained responses, effectively aligning coarse semantics with spatial details for robust and discriminative feature representation.

Unlike conventional dynamic convolution methods that generate input-dependent kernels, the proposed DCEM adopts a kernel refinement strategy, where a frozen convolution kernel is combined with a learnable convolution kernel. This design provides a stable reference for feature extraction and enables more robust optimization.

Given an input tensor $\mathbf{F}_{\text{in}} \in \mathbb{R}^{B\times C\times H\times W}$, where B, C, H, and W denote the batch size, channel number, height, and width, respectively, and the feature is fed into the DCEM. Within DCEM, the module first expands channels via a 1×1 convolution

$$\mathbf{F}_1 = \text{Conv}_{1\times 1}(\mathbf{F}_{\text{in}}) \in \mathbb{R}^{B\times(C\cdot r)\times H\times W} \tag{14}$$

where r is the channel expansion ratio that controls the dimensionality of the intermediate feature representation.

The result is batch-normalized and concatenated with its normalized version to produce

$$\mathbf{F}_2 = \text{Concat}(\mathbf{F}_1, \text{BN}(\mathbf{F}_1)) \in \mathbb{R}^{B\times 2(C\cdot r)\times H\times W} \tag{15}$$

where $\text{BN}(\cdot)$ denotes the batch normalization operation.

Next, a dynamic convolution with learnable kernel enhancement is applied to $\mathbf{F}_2$ to generate the intermediate DCEM feature $\mathbf{F}_d$

$$\mathbf{F}_d = \mathbf{F}_2 * (\mathbf{W}_{\text{conv}} + \Delta\mathbf{W}) \tag{16}$$

where $\mathbf{W}_{\text{conv}}$ is the frozen convolution kernel randomly initialized using the default PyTorch initialization scheme, kept nontrainable throughout training and not pretrained on any external datasets, while $\Delta\mathbf{W}$ represents the learnable convolution kernel for adaptive enhancement.

Furthermore, unlike conventional dynamic or adaptive convolution methods that rely on input-dependent kernel generation, DCEM does not require input-conditioned kernels. Instead, it refines the frozen kernel via the learnable convolution kernel, enhancing representation capacity while maintaining computational efficiency.

The global attention map $\mathbf{G}$ is obtained by global average pooling over $\mathbf{F}_d$ followed by a Sigmoid activation, which highlights informative channels:

$$\mathbf{G} = \text{Sigmoid}(\text{GAP}(\mathbf{F}_d)). \tag{17}$$

The feature map is first modulated by channel-wise global attention obtained via global average pooling, which guides rather than directly generates local features. To extract position-sensitive local cues, a max operation along the channel dimension is then applied to form a single-channel spatial descriptor, capturing the most salient spatial responses

$$\mathbf{M} = \max_c (\mathbf{G} \odot \mathbf{F}_{\text{in}}) \tag{18}$$

where $\odot$ denotes element-wise multiplication.

The resulting map $\mathbf{M}$ encodes salient spatial responses guided by global attention and is subsequently fed into a DCEM, followed by a Sigmoid activation function, to generate the local attention map

$$\mathbf{L} = \text{Sigmoid}(\text{DCEM}(\mathbf{M})). \tag{19}$$

Finally, the global and local attention maps are fused and applied to the input to yield the output

$$\mathbf{F}_{\text{out}} = (\mathbf{G} \odot \mathbf{L}) \odot \mathbf{F}_{\text{in}}. \tag{20}$$

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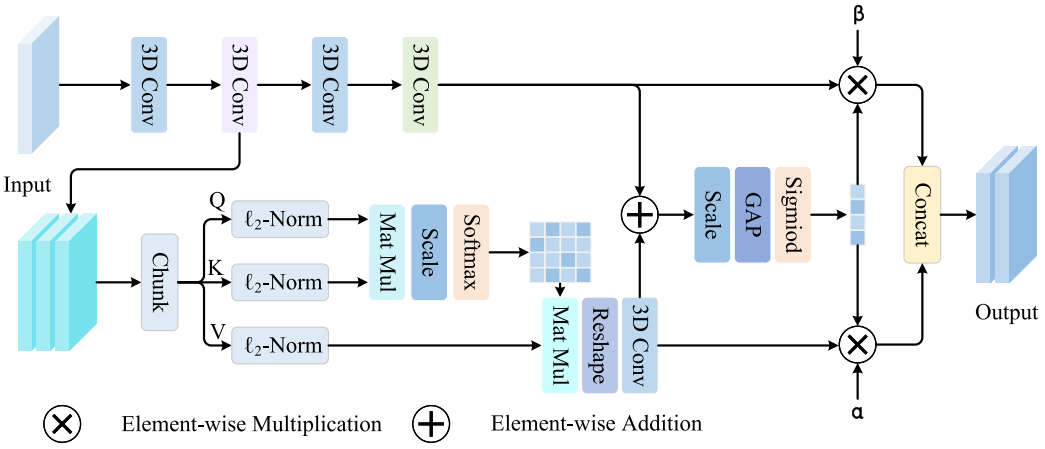


Fig. 3. Structure of the DWCAM.

D. Structure of the DWCAM

The proposed DWCAM captures both spatially extended global dependencies and fine-grained local interactions in HSI feature maps (see Fig. 3). Unlike conventional attention mechanisms that rely solely on self-attention or convolution, DWCAM integrates multihead self-attention with dynamic depthwise convolutional enhancement. It further introduces a learnable weighted fusion strategy to adaptively balance global and local responses. This hybrid design allows the module to extract complementary contextual cues while maintaining computational efficiency and representational flexibility.

The specific module implementation is as follows: let the input feature tensor be $\mathbf{H}_{in} \in \mathbb{R}^{B\times D\times H\times W}$, where B is the batch size, D is the number of channels, and H and W are the spatial dimensions. The input $\mathbf{H}_{in}$ is first expanded along the channel dimension via a 1×1 convolution and enhanced using a depthwise 3-D convolution to obtain multichannel feature maps, denoted as follows:

$$\mathbf{QKV} = \mathrm{DWConv}\left(\mathrm{Conv}_{1\times1\times1}\left(\mathbf{H}_{in}\right)\right). \tag{21}$$

For local feature modeling, $\mathbf{QKV}$ is reshaped and passed through a $1 \times 1 \times 1$ convolution followed by a grouped 3-D convolution to produce local descriptors, $\mathbf{L}_{\text{feat}} = \mathrm{DepConv}(\mathrm{Reshape}(\mathrm{Conv}_{1\times1\times1}(\mathbf{QKV})))$, where DepConv denotes a grouped convolution that captures local spatial correlations while preserving the channel-wise structure.

Simultaneously, the globally enhanced feature is obtained by splitting $\mathbf{QKV}$ into query, key, and value tensors

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} = \mathrm{Split}\left(\mathbf{QKV}\right). \tag{22}$$

The resulting tensors are subsequently reshaped into multihead representations: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{B\times H_{\text{head}}\times C_{\text{head}}\times(HW)}$, where H_{head} is the number of attention heads and $C_{head} = D/H_{head}$ is the per-head channel dimension. Queries and keys are normalized along the feature dimension to stabilize attention computation, and the attention map is computed as follows:

$$\mathbf{A} = \mathrm{Softmax}\left(\left(\mathbf{QK}^{\top}\right)\cdot\boldsymbol{\tau}\right) \tag{23}$$

where $\boldsymbol{\tau} \in \mathbb{R}^{H_{head}\times1\times1}$ is a learnable temperature parameter, which controls the sharpness of the attention distribution across different heads.

The attention-weighted values yield the globally enhanced feature

$$\mathbf{G}_{feat} = \mathbf{AV}. \tag{24}$$

The local and global features, $\mathbf{L}_{feat}$ and $\mathbf{G}_{feat}$, are then scaled by a learned channel-wise Sigmoid factor derived from global average pooling

$$\mathbf{S} = \mathrm{Sigmoid}\left(\mathrm{GAP}(\mathbf{L}_{\text{feat}} + \mathbf{G}_{\text{feat}})\right) \tag{25}$$

$$\mathbf{L}' = \mathbf{L}_{\text{feat}} \odot \mathbf{S} \tag{26}$$

$$\mathbf{G}' = \mathbf{G}_{\text{feat}} \odot \mathbf{S} \tag{27}$$

where $+$ denotes element-wise addition and $\odot$ denotes element-wise multiplication.

A dynamic weighted fusion mechanism adaptively combines these two features. Let the learnable fusion weights be $\alpha, \beta > 0$, normalized via a Softmax

$$w_1 = \frac{e^{\alpha}}{e^{\alpha}+e^{\beta}}, \quad w_2 = \frac{e^{\beta}}{e^{\alpha}+e^{\beta}} \tag{28}$$

where w_1 and w_2 are learnable normalized fusion coefficients that balance the contributions of global and local feature representations.

The final fused feature is obtained by concatenation

$$\mathbf{H}_{out} = \mathrm{Concat}\left(w_1\mathbf{G}', w_2\mathbf{L}'\right). \tag{29}$$

IV. Experiments and Discussion

This section presents a series of experimental studies using five widely used HSIs to assess the performance of the introduced FMSE-DGAF. The input HSIs are first projected into a 30-D feature space using PCA. The model is trained using the Adam optimizer with a cosine annealing learning rate scheduler, and the batch size is fixed at 64. The patch size, train-test ratio, and learning rate are treated as hyperparameters and varied according to the experimental settings. Experiments are conducted on an NVIDIA GPU with PyTorch 2.2.0 (CUDA 11.8). Following these preprocessing and training configurations, nine state-of-the-art HSI classification methods are selected for comparison, including CVSSN [41], Massformer [42], MSFIC [43], DBCT [44], SSFTT [45], SFDCT-Former [46], LSGA [47], 3DSS-Mamba [39],

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and S^{2}Mamba [40]. These comparison methods cover a range of representative HSI classification strategies, encompassing convolutional architectures, Transformer-based frameworks, and hybrid spectral–spatial feature learning schemes. The classification results are assessed quantitatively with three metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ). OA indicates the percentage of correctly classified pixels relative to the total number of pixels, while AA is the mean of per-class accuracies, providing a balanced assessment under class imbalance. The κ coefficient accounts for chance agreement between predicted labels and reference data, offering an additional measure of classification reliability. Experiments were conducted using a uniform computational platform to guarantee reproducibility and fairness. The system configuration includes an Intel Core i5-12600KF CPU, an NVIDIA RTX 4060 Ti GPU, and 64 GB of RAM.

A. Dataset Description

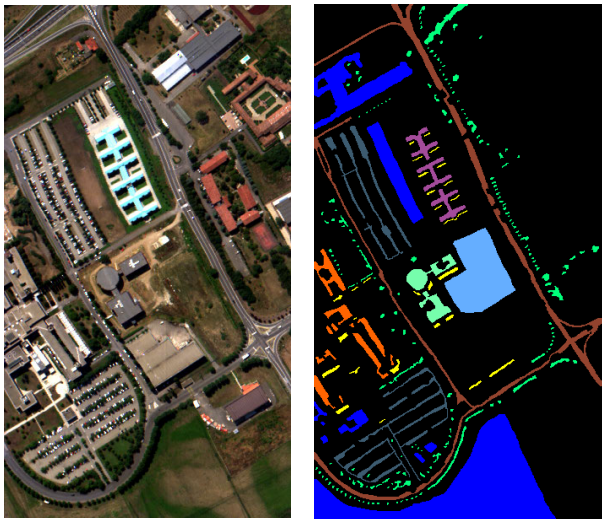
Five publicly accessible HSIs were employed for the experiments, including the Pavia University, Salinas, WHU-Hi-LongKou, WHU-Hi-HanChuan, and Botswana datasets.

The Pavia University dataset was collected over Pavia, northern Italy, using the reflective optical systems imaging spectrometer (ROSIS) on July 8, 2002. It comprises 115 spectral bands with spatial dimensions of 610 $\times$ 340 pixels. The spectral range spans 0.43–0.86 μm with a 4-nm spectral resolution, and the ground sampling distance is 1.3 m per pixel. The associated ground-truth map defines nine urban land-cover classes, totaling 42 776 labeled pixels. After removing 12 noisy bands, 103 bands were retained for classification.

The Salinas dataset was captured in 1998 over the Salinas Valley in CA, USA, utilizing the airborne visible/infrared imaging spectrometer (AVIRIS) sensor. It covers an agricultural area and contains an image of 512 $\times$ 217 pixels with 224 spectral bands, each at a spatial resolution of 3.7 m. To reduce the impact of water absorption, 20 spectral bands were removed, leaving 204 bands for subsequent analysis. The dataset provides 16 ground-truth classes representing different land cover types, with a total of 54 129 labeled pixels.

The WHU-Hi-LongKou dataset was acquired on July 17, 2018, between 13:49 and 14:37 over Longkou, Hubei, China, using a Headwall Nano-Hyperspec sensor (8-mm focal length) mounted on a DJI Matrice 600 Pro UAV. Data collection was conducted under clear, cloud-free conditions, with an ambient temperature of approximately 36 °C and relative humidity around 65%. The surveyed area represents a typical agricultural landscape, encompassing six crop types: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. HSIs were captured from an altitude of 500 m, producing frames of 550 $\times$ 400 pixels across 270 spectral bands spanning 400–1000 nm, resulting in a spatial resolution of approximately 0.463 m/pixel.

The WHU-Hi-HanChuan dataset was acquired on June 17, 2016, between 17:57 and 18:46 over Hanchuan, Hubei, China, using a Leica Aibot X6 UAV V1 equipped with a Headwall Nano-Hyperspec sensor (17 mm focal length). Data collection occurred under clear weather, with an ambient temperature of approximately 30 °C and relative humidity near 70%. The surveyed area represents a mixed urban-agricultural environment, including water bodies, built-up regions, and farmland with seven crop types. HSIs were captured from a 250 m flight altitude, producing frames of 1217 $\times$ 303 pixels across 274 spectral bands covering 400–1000 nm, resulting in a spatial resolution of roughly 0.109 m/pixel. The late-afternoon acquisition caused extensive shadowed regions due to the low solar elevation.



(a) (b)

Fig. 4. Pavia University dataset. (a) Three-band color composite image. (b) Ground-truth map.

The Botswana dataset was acquired by the NASA EO-1 satellite over the Okavango Delta from 2001 to 2004. The Hyperion sensor provides hyperspectral imagery with a spatial resolution of 30 m and 242 spectral bands covering the 400–2500-nm range. After preprocessing to remove noisy and water absorption bands, 145 bands ([10–55, 82–97, 102–119, 134–164, 187–220]) were retained for analysis. The data used in this study were collected on May 31, 2001, and include 14 land-cover classes representing wetlands and woodland environments.

For all experiments, a unified and reproducible sampling protocol was applied across the Pavia University, Salinas, WHU-Hi-LongKou, WHU-Hi-HanChuan, and Botswana datasets. Specifically, 10% of the labeled samples in each dataset were randomly selected to form the training set, while the remaining 90% were retained for independent testing. Detailed statistics of land-cover categories, including the number of samples assigned to the training and testing subsets as well as the corresponding color codes for each class, are provided in Table I. In addition, Figs. 4–8 present the false-color composite images of the five datasets together with their ground-truth annotation maps.

B. Classification Results and Analysis

Table II presents the classification performance of FMSE-DGAF and nine comparison methods on the Pavia University dataset, with the corresponding classification maps shown in Fig. 9. FMSE-DGAF achieves the best

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TABLE I

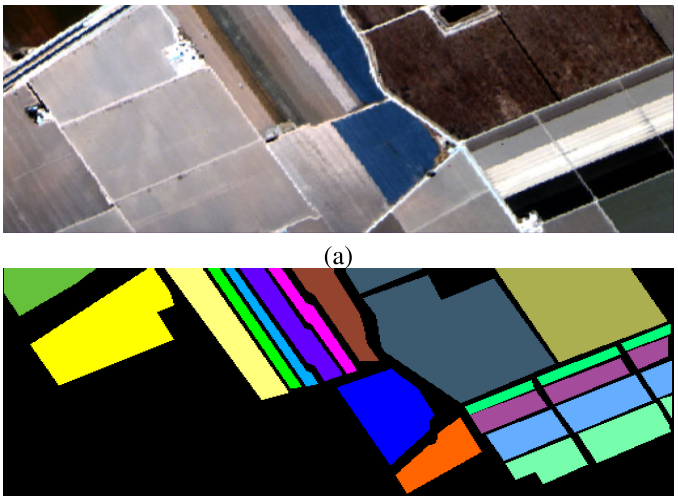
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CATEGORY IN FIVE HSI DATASETS

NO.	Color	Class	Pavia University			Salinas			WHU-Hi-LongKou			WHU-Hi-HanChuan			Botswana		
			Train	Test	Total	Train	Test	Total	Train	Test	Total	Train	Test	Total	Train	Test	Total
1		Class 1	663	5968	6631	201	1808	2009	3451	31060	34511	4473	40262	44735	27	243	270
2		Class 2	1865	16784	18649	373	3353	3726	837	7537	8374	2275	20478	22753	10	91	101
3		Class 3	210	1889	2099	198	1778	1976	303	2728	3031	1029	9258	10287	25	226	251
4		Class 4	306	2758	3064	139	1255	1394	6321	56891	63212	535	4818	5353	21	194	215
5		Class 5	134	1211	1345	268	2410	2678	415	3736	4151	120	1080	1200	27	242	269
6		Class 6	503	4526	5029	396	3563	3959	1185	10669	11854	453	4080	4533	27	242	269
7		Class 7	133	1197	1330	358	3221	3579	6706	60350	67056	590	5313	5903	26	233	259
8		Class 8	368	3314	3682	1127	10144	11271	712	6412	7124	1798	16180	17978	20	183	203
9		Class 9	95	852	947	620	5583	6203	523	4706	5229	947	8522	9469	31	283	314
10		Class 10	-	-	-	328	2950	3278	-	-	-	1052	9464	10516	25	223	248
11		Class 11	-	-	-	107	961	1068	-	-	-	1691	15220	16911	30	275	305
12		Class 12	-	-	-	193	1734	1927	-	-	-	368	3311	3679	18	164	182
13		Class 13	-	-	-	92	824	916	-	-	-	912	8204	9116	27	241	268
14		Class 14	-	-	-	107	963	1070	-	-	-	1856	16704	18560	10	85	95
15		Class 15	-	-	-	727	6541	7268	-	-	-	114	1022	1136	-	-	-
16		Class 16	-	-	-	181	1626	1807	-	-	-	7540	67861	75401	-	-	-
-		Total	4277	38499	42776	25753	231777	257530	20454	184088	204542	25753	231777	257530	324	2925	3249

TABLE II

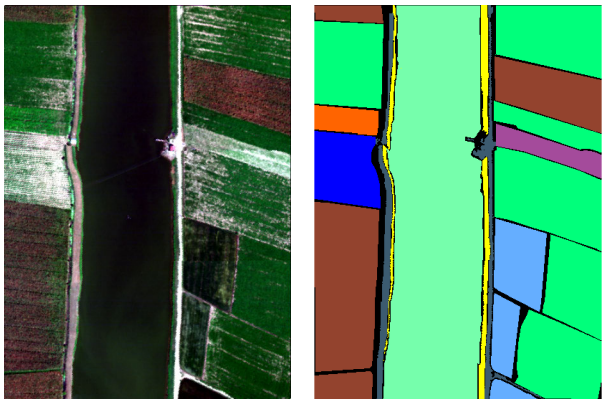
CLASS-WISE ACCURACY (%), OA (%), AA (%), AND κ (%) OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET

NO.	CVSSN	Massformer	MSFIC	DBCT	SSFTT	SFDCT-Former	LSGA	3DSS-Mamba	S^2Mamba	FMSE-DGAF
1	96.05	98.57	98.79	98.62	98.54	98.79	99.62	99.09	99.66	**99.98**
2	99.88	99.14	98.99	99.29	99.74	99.91	99.95	99.95	**99.99**	**99.99**
3	93.45	92.86	94.18	96.93	97.34	97.77	99.58	92.99	98.06	**100.0**
4	99.11	98.69	99.03	99.39	99.34	99.10	98.28	**99.78**	98.53	98.96
5	99.61	**100.0**	99.98	99.96	**100.0**	**100.0**	99.51	**100.0**	99.75	**100.0**
6	98.61	98.67	99.37	99.32	99.74	99.89	**100.0**	99.43	**100.0**	**100.0**
7	96.23	99.60	99.37	99.08	99.68	**100.0**	99.50	99.14	**100.0**	**100.0**
8	91.63	95.20	94.89	91.79	96.02	93.75	98.95	92.02	97.52	**99.67**
9	96.63	99.78	**100.0**	99.68	**100.0**	**100.0**	99.39	98.95	**100.0**	**100.0**
OA (%)	97.86	98.37	98.49	98.44	99.10	99.04	99.64	98.65	99.52	**99.89**
AA (%)	96.80	98.06	98.29	98.23	98.93	98.80	99.42	97.93	99.28	**99.84**
κ (%)	97.16	97.84	97.99	97.93	98.81	98.73	99.52	98.22	99.36	**99.86**



(a) (b)

Fig. 5. Salinas dataset. (a) Three-band color composite image. (b) Ground-truth map.



(a) (b)

Fig. 6. WHU-Hi-LongKou dataset. (a) Three-band color composite image. (b) Ground-truth map.

performance, with an OA of 99.89%, an AA of 99.84%, and a κ of 99.86%, maintaining high accuracy across all classes, particularly excelling in classes with complex spectral distributions and irregular boundaries (e.g., Classes 1, 3, and 8). LSGA and S^2Mamba achieve OAs of 99.64% and 99.52%, respectively, demonstrating strong competitiveness among the existing approaches. The recently proposed 3DSS-Mamba attains an OA of 98.65%, while SSFTT and SFDCT-Former achieve OAs of 99.10% and 99.04%, with slightly

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TABLE III
CLASS-WISE ACCURACY (%), OA (%), AA (%), AND κ (%) ACHIEVED BY DIFFERENT METHODS ON THE SALINAS DATASET

NO.	CVSSN	Massformer	MSFIC	DBCT	SSFTT	SFDCT-Former	LSGA	3DSS-Mamba	S^{2}Mamba	FMSE-DGAF
1	**100.0**	**100.0**	99.95	**100.0**	**100.0**	**100.0**	99.95	**100.0**	**100.0**	**100.0**
2	99.95	99.97	**100.0**	**100.0**	**100.0**	**100.0**	99.94	**100.0**	**100.0**	**100.0**
3	99.85	**100.0**	99.46	**100.0**	99.95	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
4	99.78	98.67	98.42	99.60	**100.0**	**100.0**	99.40	99.85	**100.0**	**100.0**
5	99.65	99.80	99.68	99.63	99.68	99.88	99.96	99.92	99.88	**100.0**
6	**100.0**	**100.0**	99.89	**100.0**	99.89	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
7	**100.0**	99.71	99.94	99.96	99.94	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
8	95.97	97.07	96.71	98.31	95.31	98.86	99.00	99.72	98.84	**99.88**
9	99.30	99.73	99.76	99.79	99.02	99.95	99.87	99.95	99.84	**99.98**
10	99.43	99.49	99.51	99.59	99.59	99.87	99.65	99.73	99.93	**100.0**
11	98.30	99.93	98.92	98.97	98.92	**100.0**	99.61	99.90	**100.0**	**100.0**
12	99.63	99.31	99.68	99.66	**100.0**	**100.0**	99.89	**100.0**	99.88	**100.0**
13	**100.0**	99.32	99.78	99.76	99.77	99.89	**100.0**	98.57	**100.0**	**100.0**
14	98.58	98.83	99.81	99.58	98.71	**100.0**	98.58	99.37	99.79	**100.0**
15	96.73	94.46	95.38	97.93	97.09	97.62	95.96	95.08	99.54	**100.0**
16	**100.0**	**100.0**	99.92	**100.0**	99.98	**100.0**	99.32	**100.0**	99.63	**100.0**
OA (%)	98.47	98.45	98.23	99.25	98.72	99.42	99.15	99.18	99.64	**99.96**
AA (%)	99.16	99.14	99.25	99.55	99.24	99.75	99.52	99.50	99.83	**99.93**
κ (%)	98.30	98.27	98.03	99.16	98.58	99.36	99.05	99.08	99.60	**99.95**

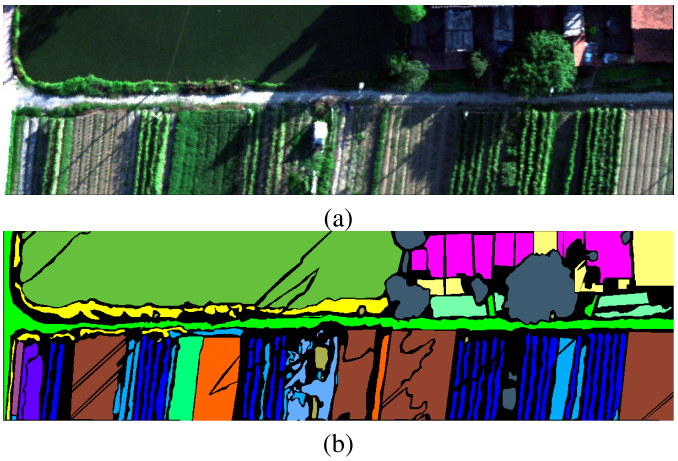


Fig. 7. WHU-Hi-HanChuan dataset. (a) Three-band color composite image. (b) Ground-truth map.

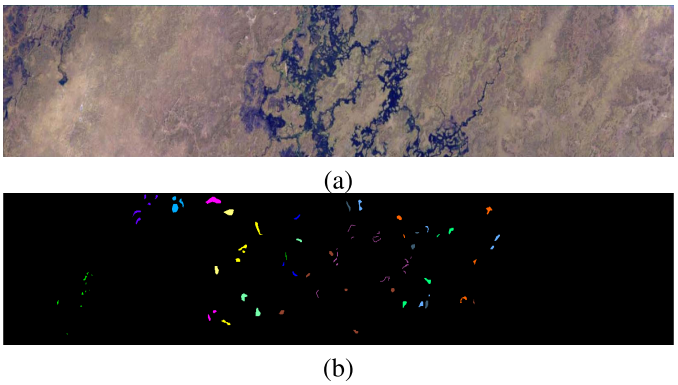


Fig. 8. Botswana dataset. (a) Three-band color composite image. (b) Ground-truth map.

lower accuracy in certain spectrally similar classes. CVSSN and Massformer attain OAs of 97.86% and 98.37%, respectively, and exhibit lower class-wise accuracy in categories with complex spectral characteristics, highlighting their limitations in capturing discriminative spectral–spatial features.

The classification performance of FMSE-DGAF and nine comparison algorithms on the Salinas dataset is summarized in Table III. A visual illustration of the corresponding results is provided in Fig. 10. As shown, FMSE-DGAF achieves outstanding performance, delivering an OA of 99.96%, an AA of 99.93%, and a κ of 99.95%, which demonstrates its strong discriminative capability on this large-scale and spectrally complex dataset. From the class-wise accuracy results, FMSE-DGAF achieves 100% classification accuracy in 12 out of 16 categories, indicating remarkable consistency and stability. It particularly excels in classes with high spectral similarity and fine-grained textures (e.g., Classes 8, 15, and 16). While S^{2}Mamba and SFDCT-Former also show strong performance, achieving OAs of 99.64% and 99.42%, slight accuracy reductions are observed in more challenging classes. 3DSS-Mamba, DBCT, and LSGA achieve OAs of 99.18%, 99.25%, and 99.15%, respectively. CVSSN, Massformer, and MSFIC achieve comparatively lower overall accuracies, highlighting FMSE-DGAF's advantage in handling large-scale hyperspectral scenes with complex spectral–spatial patterns.

Table IV presents a quantitative comparison of ten competing approaches on the WHU-Hi-LongKou dataset, while the associated thematic maps are illustrated in Fig. 11. The proposed FMSE-DGAF consistently delivers superior results, achieving an OA of 99.91%, an AA of 99.76%, and a κ of 99.89%. A category-level evaluation further reveals that FMSE-DGAF attains the leading or highly competitive performance across the majority of land-cover classes, demonstrating clear advantages in scenarios characterized by intricate spatial layouts and subtle spectral distinctions, such as Classes 2 and 5. In comparison, S^{2}Mamba achieves a competitive OA of 99.81%, closely following the proposed method. SFDCT-Former, 3DSS-Mamba, LSGA, and DBCT report overall accuracies of 99.75%, 99.43%, 99.28%, and 99.22%, respectively, and tend to exhibit reduced effectiveness

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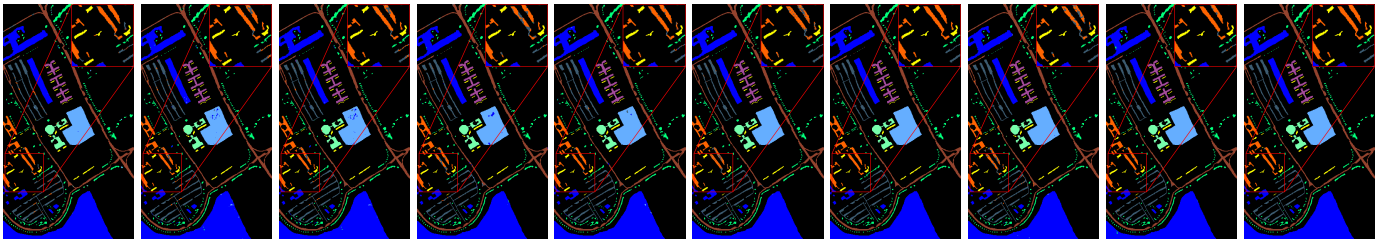


Fig. 9. Comparative visualization of classification results from different approaches on the Pavia University dataset, based on 10% of the training samples. (a) CVSSN (OA = 97.86%). (b) Massformer (OA = 98.37%). (c) MSFIC (OA = 98.49%). (d) DBCT (OA = 98.44%). (e) SSFTT (OA = 99.10%). (f) SFDCT-Former (OA = 99.04%). (g) LSGA (OA = 99.64%). (h) 3DSS-Mamba (OA = 98.65%). (i) S^{2}Mamba (OA = 99.52%). (j) FMSE-DGAF (OA = 99.89%).

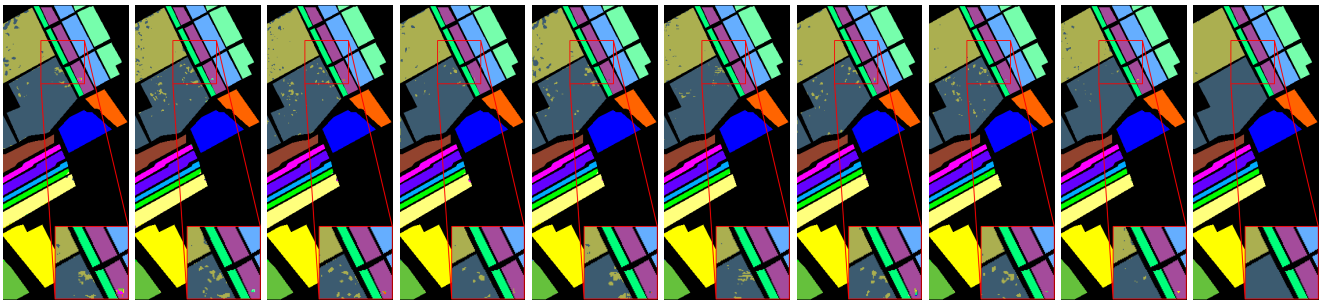


Fig. 10. Comparative visualization of classification results from different approaches on the Salinas dataset, based on 10% of the training samples. (a) CVSSN (OA = 98.47%). (b) Massformer (OA = 98.45%). (c) MSFIC (OA = 98.23%). (d) DBCT (OA = 99.25%). (e) SSFTT (OA = 98.72%). (f) SFDCT-Former (OA = 99.42%). (g) LSGA (OA = 99.15%). (h) 3DSS-Mamba (OA = 99.18%). (i) S^{2}Mamba (OA = 99.64%). (j) FMSE-DGAF (OA = 99.96%).

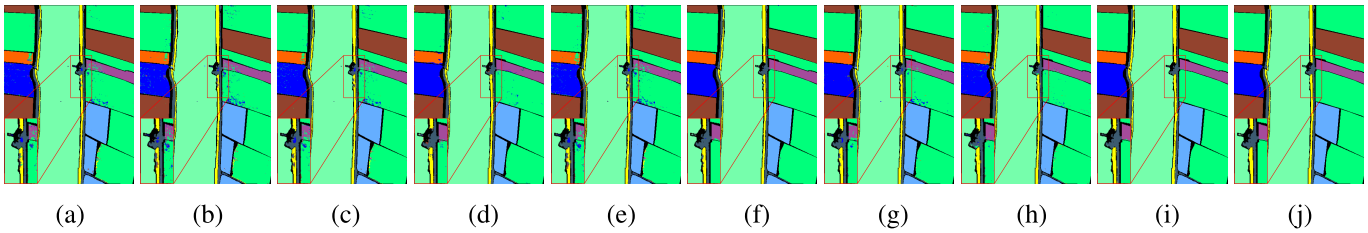


Fig. 11. Comparative visualization of classification results from different approaches on the WHU-Hi-LongKou dataset, based on 10% of the training samples. (a) CVSSN (OA = 98.82%). (b) Massformer (OA = 98.53%). (c) MSFIC (OA = 98.86%). (d) DBCT (OA = 99.22%). (e) SSFTT (OA = 98.64%). (f) SFDCT-Former (OA = 99.75%). (g) LSGA (OA = 99.28%). (h) 3DSS-Mamba (OA = 99.43%). (i) S^{2}Mamba (OA = 99.81%). (j) FMSE-DGAF (OA = 99.91%).

TABLE IV

CLASS-WISE ACCURACY (%), OA (%), AA (%), AND κ (%) OBTAINED BY DIFFERENT METHODS ON THE WHU-HI-LONGKOU DATASET

NO.	CVSSN	Massformer	MSFIC	DBCT	SSFTT	SFDCT-Former	LSGA	3DSS-Mamba	S^{2}Mamba	FMSE-DGAF
1	99.40	99.23	99.43	99.82	99.24	99.98	99.74	99.59	99.89	**99.99**
2	94.47	91.42	93.58	94.76	92.54	99.53	95.76	97.86	99.41	**99.81**
3	98.84	97.18	99.39	99.92	98.89	**100.0**	99.25	98.59	99.76	**100.0**
4	98.80	98.97	98.76	99.82	98.53	99.84	99.46	99.49	99.91	**99.97**
5	94.16	93.63	95.11	97.33	95.11	99.53	99.02	94.85	99.17	**100.0**
6	99.66	98.88	99.89	99.67	99.49	99.91	99.63	99.82	99.97	**100.0**
7	99.98	99.96	99.98	99.97	99.94	99.96	99.98	**99.99**	99.98	99.98
8	94.59	92.49	94.88	94.13	94.45	96.26	94.56	98.64	97.97	**98.74**
9	95.02	95.49	96.70	93.04	95.86	**99.45**	96.91	97.37	99.43	99.36
OA (%)	98.82	98.53	98.86	99.22	98.64	99.75	99.28	99.43	99.81	**99.91**
AA (%)	97.21	96.35	97.52	97.61	97.08	99.38	98.26	99.46	99.49	**99.76**
κ (%)	98.45	98.07	98.51	98.98	98.21	99.67	99.05	99.24	99.75	**99.89**

in classes affected by spectral resemblance or irregular object boundaries. The remaining methods, including CVSSN, Massformer, and SSFTT, achieve relatively lower overall scores, indicating weaker capability in modeling fine-scale spectral–spatial information. Collectively, these observations demonstrate that FMSE-DGAF provides a more

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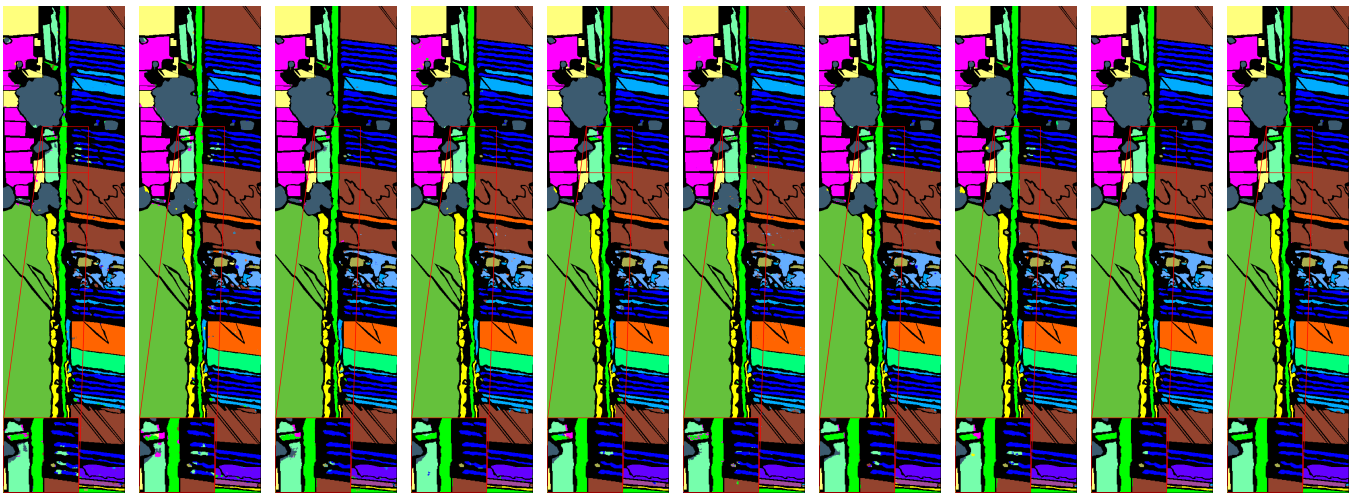


Fig. 12. Comparative visualization of classification results from different approaches on the WHU-Hi-HanChuan dataset, based on 10% of the training samples. (a) CVSSN (OA = 98.50%). (b) Massformer (OA = 98.21%). (c) MSFIC (OA = 98.93%). (d) DBCT (OA = 99.25%). (e) SSFTT (OA = 98.92%). (f) SFDCT-Former (OA = 99.03%). (g) LSGA (OA = 98.98%). (h) 3DSS-Mamba (OA = 98.88%). (i) S^{2}Mamba (OA = 99.64%). (j) FMSE-DGAF (OA = 99.89%).

TABLE V

CLASS-WISE ACCURACY (%), OA (%), AA (%), AND κ (%) ACHIEVED BY DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET

NO.	CVSSN	Massformer	MSFIC	DBCT	SSFTT	SFDCT-Former	LSGA	3DSS-Mamba	S^{2}Mamba	FMSE-DGAF
1	97.45	96.84	97.87	99.07	98.69	98.82	98.39	98.71	99.70	**99.88**
2	98.49	98.50	98.98	99.45	98.82	99.22	99.15	99.45	99.73	**100.0**
3	99.26	98.56	99.78	98.66	99.29	98.14	99.42	99.15	99.86	**99.98**
4	99.63	99.62	99.75	99.07	99.77	99.88	99.69	98.53	99.92	**100.0**
5	98.66	97.84	**100.0**	99.91	91.82	99.63	99.72	99.40	97.32	99.36
6	98.54	95.95	96.11	97.85	93.43	95.27	95.08	94.87	98.06	**99.12**
7	93.14	93.82	98.98	98.07	98.38	96.97	97.62	97.39	99.20	**99.81**
8	97.94	98.08	98.29	98.69	98.08	98.32	98.57	98.51	99.74	**99.78**
9	97.61	96.90	98.43	98.66	97.92	98.60	97.74	95.69	99.11	**99.79**
10	99.58	98.93	99.50	99.80	99.01	99.81	99.84	99.44	99.33	**99.86**
11	97.93	96.70	98.43	99.57	98.75	98.79	98.61	98.74	99.72	**99.94**
12	96.07	96.76	99.62	99.82	98.60	98.68	97.47	99.50	**100.0**	99.91
13	96.43	96.26	97.68	97.52	99.33	96.86	97.18	95.81	98.57	**99.67**
14	98.97	98.33	98.80	98.87	98.70	99.11	99.25	98.92	99.53	**99.87**
15	92.69	99.37	97.87	99.01	94.64	98.13	**100.0**	94.01	95.06	99.80
16	99.95	99.96	99.93	99.88	99.90	99.95	99.94	**99.99**	99.98	99.97
OA(%)	98.50	98.21	98.93	99.25	98.92	99.03	98.98	98.88	99.64	**99.89**
AA(%)	97.65	97.66	98.75	98.63	97.82	98.51	98.60	98.01	99.05	**99.79**
κ (%)	98.25	97.90	98.75	99.12	98.73	98.86	98.81	98.69	99.58	**99.87**

dependable representation of joint spectral–spatial characteristics and maintains consistent classification performance.

Table V summarizes the comparative results of ten approaches evaluated on the WHU-Hi-HanChuan, while the related prediction maps are provided in Fig. 12. FMSE-DGAF continues to exhibit superior performance, delivering an OA of 99.89%, an AA of 99.79%, and a κ of 99.87%, exceeding all competing methods. S^{2}Mamba achieves the second-best performance with an OA of 99.64%. DBCT, SSFTT, and SFDCT-Former achieve OAs of 99.25%, 98.92%, and 99.03%, respectively, while the newly added 3DSS-Mamba attains an OA of 98.88%, with slightly lower accuracy in some spectrally similar classes (e.g., Classes 5 and 6). Massformer attains an overall OA of 98.21%, reflecting its reduced effectiveness in modeling fine-grained spectral–spatial details. FMSE-DGAF maintains high accuracy, with 12 out of 16 classes achieving the highest or near-highest accuracy, demonstrating its robustness to spectral–spatial variations and its ability to discriminate complex land-cover categories. Overall, these results confirm the efficiency and reliability of FMSE-DGAF in HSI classification tasks.

To further evaluate the generalization capability and scalability of the proposed FMSE-DGAF, a supplementary experiment was conducted on the Botswana dataset. Compared to the previous four datasets, the Botswana dataset exhibits notable differences in its characteristics. It has a lower spatial resolution (30 m) and mainly consists of natural scenes such as wetlands. In contrast, Pavia University primarily represents urban and campus environments, Salinas corresponds to agricultural regions, and the WHU-Hi series

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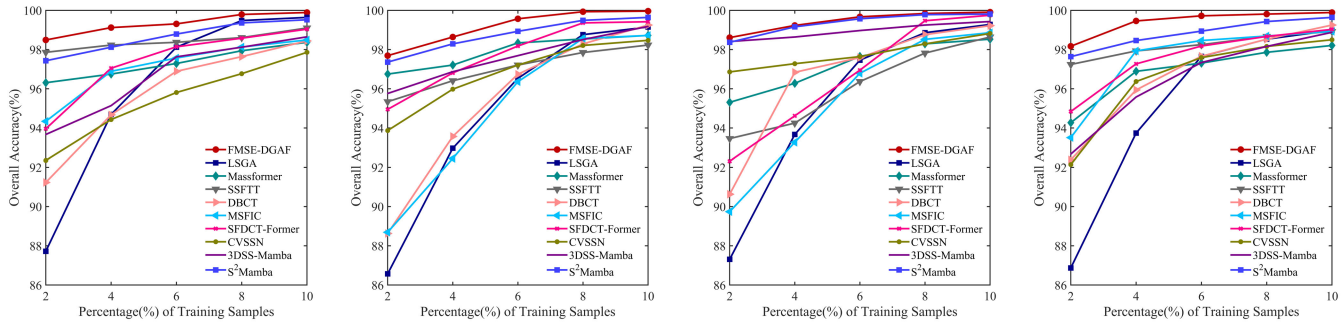


Fig. 13. Evaluation of OA under different training sample ratios. (a) Pavia University. (b) Salinas. (c) WHU-Hi-LongKou. (d) WHU-Hi-HanChuan datasets.

TABLE VI

CLASSIFICATION PERFORMANCE COMPARISON OF FMSE-DGAF AND REPRESENTATIVE BASELINES ON THE BOTSWANA DATASET

Methods	OA (%)	AA (%)	κ (%)
SFDCT-Former	98.49	98.53	98.37
S^{2}Mamba	99.21	99.27	99.15
FMSE-DGAF	**99.76**	**99.78**	**99.74**

covers complex mixed scenes of urban areas and farmland, all with higher spatial resolutions (e.g., 0.109 m in the WHU-Hi-HanChuan dataset). Notably, the spatial resolution difference among these datasets is close to 300×, posing a significant challenge to model generalization. Considering the overall performance of the compared methods on the aforementioned datasets, two representative strong baselines, S^{2}Mamba and SFDCT-Former, were selected for this evaluation. As shown in Table VI, FMSE-DGAF still achieves the best performance, with OA, AA, and κ reaching 99.76%, 99.78%, and 99.74%, respectively. In comparison, S^{2}Mamba achieves 99.21%, 99.27%, and 99.15%, while SFDCT-Former achieves 98.49%, 98.53%, and 98.37%, respectively. These results demonstrate that FMSE-DGAF is robust across a 300× spatial resolution difference, consistently achieving excellent classification performance across datasets with significantly different characteristics, further confirming its effectiveness and strong generalization capability across diverse hyperspectral scenarios.

C. Parameter Analysis

1) Analysis of Training Sample Ratio Effects on OA: Fig. 13 illustrates the classification performance of ten representative methods under different training sample ratios ranging from 2% to 10% on four benchmark HSIs. Across all experimental settings, the proposed FMSE-DGAF consistently maintains superior accuracy compared with the competing approaches. This consistent advantage under limited and relatively abundant training samples indicates that FMSE-DGAF is less sensitive to variations in training data availability and exhibits strong generalization capability and robustness.

On the Pavia University dataset, FMSE-DGAF attains an OA of 98.49% when trained with only 2% of labeled samples, and the accuracy gradually increases to 99.89% as the training proportion rises to 10%. Among the competitors, SSFTT demonstrates relatively strong robustness, achieving 97.86% OA with 2% training samples, followed by S^{2}Mamba (97.43%). In comparison, LSGA and DBCT attain 87.72% and 91.23% at 2%, respectively, showing a stronger reliance on larger training sets. Even as the training proportion increases, FMSE-DGAF maintains a clear advantage, outperforming all competitors by 1%–2% at higher proportions.

On the Salinas dataset, FMSE-DGAF starts at 97.69% OA with 2% training samples and reaches 99.96% at 10%, achieving near-perfect classification. Among the competing models, S^{2}Mamba and Massformer demonstrate strong performance under small training sample conditions, achieving 97.36% and 96.75% at 2%, respectively. In comparison, LSGA, MSFIC, and DBCT show lower accuracy at 2% sample sizes (86.57%, 88.69%, and 88.63%), demonstrating FMSE-DGAF's effectiveness under different labeled data conditions.

For the WHU-Hi-LongKou dataset, FMSE-DGAF consistently leads across all training proportions, achieving 98.61% with 2% of training samples and increasing to 99.91% at 10%. Notably, S^{2}Mamba and 3DSS-Mamba exhibit exceptional performance, achieving 98.41% and 98.37% at 2%, respectively, which are very close to the proposed method. Models such as LSGA and MSFIC perform relatively poorly at low training sizes; for example, LSGA reaches only 87.31% at 2%, indicating limited feature extraction capability. Although the accuracy of other algorithms improves as the training proportion increases, FMSE-DGAF maintains the highest performance throughout.

Finally, on the WHU-Hi-HanChuan dataset, FMSE-DGAF obtains 98.17% OA with 2% training samples and 99.89% at 10%, consistently outperforming all competing methods. Among the comparison approaches, S^{2}Mamba obtains the most competitive performance, achieving 97.64% at 2% samples and reaching 99.64% at 10% samples, making it the closest competitor to FMSE-DGAF. SSFTT also exhibits stable and strong performance across different training ratios, maintaining OA values above 97% even under limited training conditions. In contrast, LSGA shows the lowest accuracy at the smallest training ratio (86.87%), indicating relatively weaker generalization ability under extremely limited labeled

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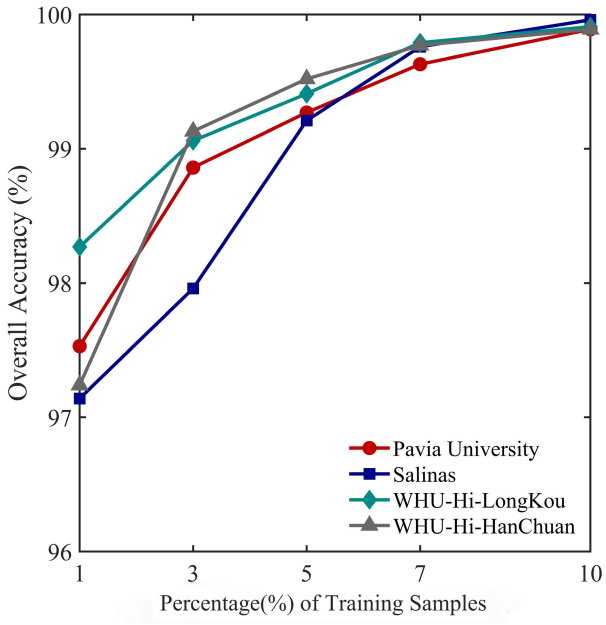


Fig. 14. OA of FMSE-DGAF with 1%, 3%, 5%, 7%, and 10% training samples on the Pavia University, Salinas, WHU-Hi-LongKou, and WHU-Hi-HanChuan datasets.

data. Meanwhile, 3DSS-Mamba achieves relatively modest performance at low training ratios (92.67% at 2% samples), but improves significantly as the number of training samples increases, suggesting a higher dependence on sufficient training data.

Overall, these results demonstrate FMSE-DGAF's superior classification accuracy, stability, and generalization across different datasets and training sample ratios.

To further evaluate the robustness of FMSE-DGAF in extreme data-scarce scenarios, we conducted additional experiments using training sample ratios of 1%, 3%, 5%, 7%, and 10% per class on all four benchmark datasets. The results are visualized in Fig. 14. With only 1% of labeled samples, FMSE-DGAF achieves an OA of 97.53% on Pavia University, 97.14% on Salinas, 98.27% on WHU-Hi-LongKou, and 97.24% on WHU-Hi-HanChuan. These results indicate that the model maintains competitive performance even with extremely limited labeled data.

The learning curves in Fig. 14 show rapid performance improvements as the training ratio increases. When the ratio rises from 1% to 3%, the OA increases by 1.33% on Pavia University (reaching 98.86%) and by 1.89% on WHU-Hi-HanChuan (reaching 99.13%). With 5% of samples, the OA exceeds 99% on all datasets. At 10%, the accuracies reach 99.89% for Pavia University, 99.96% for Salinas, 99.91% for WHU-Hi-LongKou, and 99.89% for WHU-Hi-HanChuan.

In summary, these results demonstrate that FMSE-DGAF maintains stable performance with minimal training data and converges quickly as more samples become available. This demonstrates its practical potential for HSI classification tasks where annotated data are limited.

2) Analysis of Learning Rate Effects on OA, AA, and \kappa: In this study, the impact of learning rates set in the range of 0.001 to 0.005 on FMSE-DGAF was evaluated across four datasets, using OA, AA, and κ as performance metrics, with results obtained from five repeated experiments and averaged to ensure stable performance. Fig. 15 indicates that Pavia University achieved its highest performance at a learning rate of 0.004, with OA = 99.89%, AA = 99.84%, and κ = 99.86%. For Salinas, the optimal learning rate was 0.003, yielding OA = 99.96%, AA = 99.93%, and κ = 99.95%. The WHU-Hi-LongKou dataset performed best at 0.002, with OA = 99.91%, AA = 99.76%, and κ = 99.89%. On WHU-Hi-HanChuan, the model reached its optimal performance when the learning rate was set to 0.004, achieving OA = 99.89%, AA = 99.79%, and κ = 99.87%. The experimental results demonstrate that the choice of learning rate has a significant impact on the classification performance of the model. Both excessively large and excessively small learning rates lead to performance degradation, whereas moderate learning rates consistently achieve superior classification results across different datasets. This can be attributed to the fact that a large learning rate may cause the optimization process to overshoot the optimal region, resulting in oscillations during training and reduced optimization effectiveness. In contrast, an overly small learning rate restricts the magnitude of parameter updates, making it difficult for the model to reach a sufficiently optimized state within the predefined number of training epochs, thereby affecting the final classification performance. Therefore, an appropriate learning rate is essential for effectively learning discriminative features from hyperspectral data. Overall, learning rates within the range of 0.002–0.004 consistently exhibit strong adaptability and competitive performance across the evaluated datasets. Consequently, when applying FMSE-DGAF to new hyperspectral datasets, this range can be regarded as a reasonable initial search interval and further refined according to the characteristics of complex hyperspectral remote sensing scenes.

3) Analysis of Patch Size Effects on OA, AA, and κ: The classification accuracy of FMSE-DGAF varies significantly with different patch sizes. Fig. 16 illustrates the variations in the average OA, AA, and κ over five independent runs as the patch size ranges from 9×9 to 25×25 on these datasets used in our experiments. Specifically, the optimal patch sizes are 19×19 for the Pavia University and WHU-Hi-HanChuan datasets, 17×17 for the Salinas dataset, and 15×15 for the WHU-Hi-LongKou dataset. From the overall trend, it can be observed that across all four datasets, as the patch size increases from small to moderate scales, OA, AA, and κ steadily improve, indicating that an enlarged spatial receptive field facilitates more discriminative feature representation. However, further increasing the patch size results in a noticeable performance degradation, suggesting that excessively large patches introduce redundant spatial information and weaken the classification capability. In practical applications, the patch size should be selected by balancing local detail preservation and global contextual modeling. A moderate range of patch sizes is recommended as the first choice, followed by fine-tuning around this range to achieve optimal performance.

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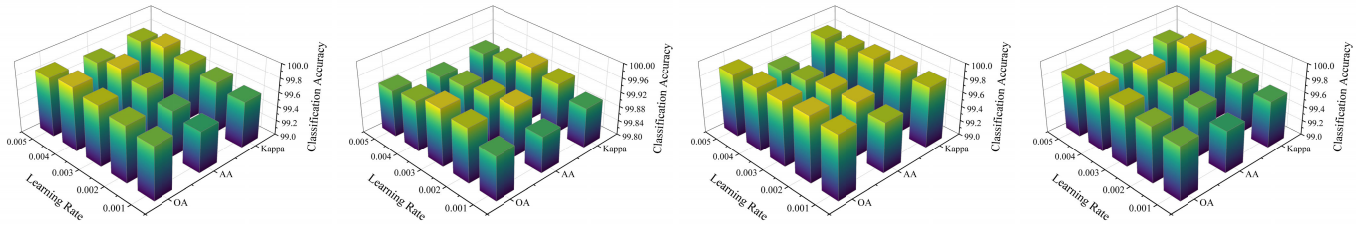


Fig. 15. Evaluation of quantitative metrics under different learning rates. (a) Pavia University. (b) Salinas. (c) WHU-Hi-LongKou. (d) WHU-Hi-HanChuan datasets.

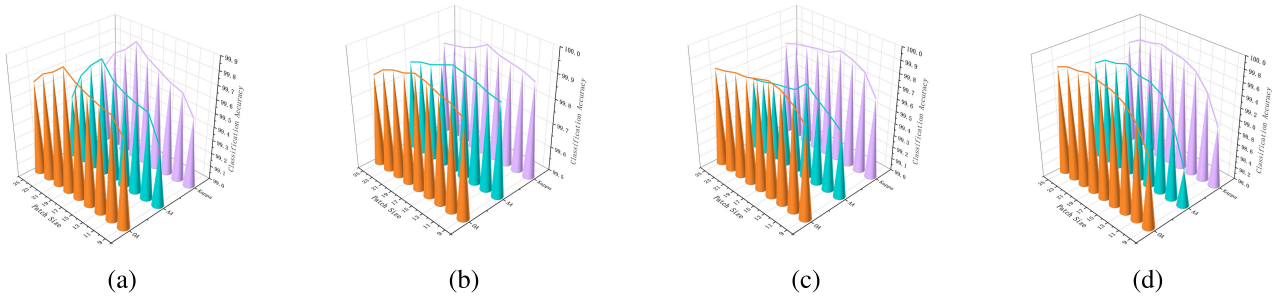


Fig. 16. Evaluation of quantitative metrics under different patch sizes. (a) Pavia University. (b) Salinas. (c) WHU-Hi-LongKou. (d) WHU-Hi-HanChuan datasets.

TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON ON FOUR HYPERSPECTRAL DATASETS

Methods	Pavia University			Salinas			WHU-Hi-LongKou			WHU-Hi-HanChuan		
	Training (m)	Testing (s)	FLOPs	Training (m)	Testing (s)	FLOPs	Training (m)	Testing (s)	FLOPs	Training (m)	Testing (s)	FLOPs
CVSSN	1.62	1.43	76.44M	2.59	2.67	104.35M	13.63	25.58	138.67M	14.58	28.21	147.92M
Massformer	1.26	2.35	75.86M	1.76	2.57	99.46M	12.87	23.47	121.72M	13.78	25.91	129.88M
MSFIC	1.17	1.34	73.29M	**1.32**	2.41	96.37M	12.43	22.16	128.37M	13.36	24.53	136.75M
DBCT	**0.92**	1.27	67.83M	1.56	**1.43**	77.71M	10.96	21.69	95.42M	11.72	23.98	101.63M
SSFTT	0.95	1.15	**64.37M**	1.28	1.96	**75.31M**	11.64	25.76	**93.25M**	12.41	28.32	**99.21M**
SFDCT-Former	1.34	2.76	78.52M	1.72	2.31	84.12M	13.89	27.29	130.58M	14.96	30.11	139.05M
LSGA	2.57	3.25	147.65M	4.01	3.12	154.63M	15.46	29.54	172.17M	16.82	32.67	183.41M
3DSS-Mamba	3.71	1.96	141.63M	4.15	3.64	167.49M	13.79	26.32	206.74M	15.76	31.98	226.49M
S^2Mamba	1.57	1.46	85.64M	1.97	2.64	121.47M	12.15	22.98	132.32M	13.62	29.74	148.85M
FMSE-DGAF	1.35	**1.19**	69.78M	1.89	2.17	89.45M	**10.21**	**20.35**	118.97M	**11.02**	**22.87**	126.08M

D. Computational Complexity Analysis

To quantitatively assess computational complexity, Table VII reports the training time, testing time, and FLOPs of all methods across four hyperspectral datasets.

For CNN- and Transformer-based approaches, Lightweight models, including SSFTT and DBCT, are characterized by relatively low computational costs across all evaluated datasets. Specifically, SSFTT achieves the lowest FLOPs on all four datasets (64.37, 75.31, 93.25, and 99.21 M), while DBCT consistently maintains the shortest or near-shortest training and testing times (e.g., 0.92-min training time on Pavia University and 1.43-s testing time on Salinas), demonstrating excellent computational efficiency. Similarly, MSFIC and Massformer also maintain moderate computational costs, with FLOPs generally below 140 M and stable runtime performance across different datasets. In contrast, more complex models such as LSGA and SFDCT-Former exhibit significantly higher computational overhead. For instance, LSGA reaches up to 183.41 M FLOPs on the WHU-Hi-HanChuan dataset and consistently incurs longer training and testing times, reflecting the increased computational burden introduced by its complex attention mechanisms and large-scale feature modeling.

The Mamba-based methods (3DSS-Mamba and S^2Mamba) are designed to alleviate the quadratic complexity of Transformer architectures through linear-complexity state space modeling. However, as shown in Table VII, their practical computational cost remains relatively high. In particular, 3DSS-Mamba exhibits the highest FLOPs among all compared methods, reaching 226.49 M on the WHU-Hi-HanChuan dataset, along with consistently longer training time (e.g., 3.71 min on Pavia University and 4.15 min on Salinas). In comparison, S^2Mamba achieves a certain reduction in computational cost, but still requires higher FLOPs and nonnegligible runtime, indicating that Mamba-based methods may still incur considerable resource consumption in HSI classification tasks despite their theoretical efficiency advantages.

In comparison, the proposed FMSE-DGAF demonstrates an effective tradeoff between computational efficiency and classification performance. As shown in Table VII, FMSE-DGAF consistently achieves competitive or superior

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runtime performance across all datasets. For example, it obtains the shortest testing time on Pavia University (1.19 s), as well as the lowest training and testing time on the WHU-Hi-LongKou (10.21 min/20.35 s) and WHU-Hi-HanChuan (11.02 min/22.87 s) datasets. Meanwhile, its FLOPs remain at a relatively low level (ranging from 69.78 to 126.08 M), which is significantly lower than those of high-complexity models such as LSGA and 3DSS-Mamba. Furthermore, despite incorporating frequency-domain operations, FMSE-DGAF maintains relatively low computational complexity. Specifically, the computational complexity of the employed rFFT2 operation is approximately $\mathcal{O}(CHW\log(HW))$, where C, H, and W denote the channel number, height, and width of the feature map, respectively. This complexity originates from the FFT, whose computational cost grows only logarithmically with spatial resolution. In contrast, the complexity of a standard convolution operation is $\mathcal{O}(HWK^2C_{in}C_{out})$, where K denotes the convolution kernel size, and C_{in} and C_{out} represent the input and output channel dimensions, respectively. Compared with conventional convolutions, the proposed frequency-domain transformation avoids the quadratic dependency on kernel size (K^2), thereby reducing computational overhead under large-kernel feature modeling scenarios to a certain extent. Therefore, although frequency-domain transformations introduce additional FFT computations, their overall computational complexity remains manageable and does not impose significant additional computational burden in practice. Moreover, compared with methods based on sequential state modeling (e.g., 3DSS-Mamba), which rely on sequential state propagation or implicit global dependency modeling, FMSE-DGAF achieves global feature interaction through a single frequency-domain transformation step. This design avoids the need for iterative token-wise processing and reduces the reliance on high-dimensional feature expansion. In addition, the amplitude-guided modulation is implemented using lightweight (1×1) convolutions, which perform efficient cross-channel feature interaction without introducing additional spatial convolutional overhead. As a result, the proposed method effectively reduces redundant computations while preserving global receptive field modeling capability and maintaining relatively low computational overhead.

To further analyze the computational resource requirements, Table VIII reports the peak GPU memory consumption of FMSE-DGAF during both training and inference across four benchmark datasets, where "Train Mem" and "Infer Mem" denote the peak GPU memory usage during training and inference, respectively. As shown in Table VIII, the training memory ranges from 377.25 to 412.23 MB, while the inference memory remains between 21.44 and 54.35 MB. This relatively small variation indicates that FMSE-DGAF maintains stable memory consumption across different datasets. In addition, the significantly lower memory requirement during inference further demonstrates its efficiency in deployment scenarios. Although frequency-domain transformations are introduced in FMSEM, they do not introduce substantial additional memory overhead. This efficiency is mainly attributed to the absence of complex multibranch architectures and the use of FFT-based frequency transformations, which avoid the need for additional multiscale feature branches and large intermediate feature storage. Overall, these results confirm that FMSE-DGAF achieves a favorable balance between computational efficiency and memory footprint, making it suitable for deployment in resource-constrained environments. Combined with the superior classification results reported in Tables II–VI and Figs. 9–12, FMSE-DGAF achieves strong overall competitiveness in terms of accuracy, computational cost, and efficiency.

TABLE VIII
GPU MEMORY CONSUMPTION OF FMSE-DGAF ON DIFFERENT DATASETS

Datasets	Train Mem (MB)	Infer Mem (MB)
Pavia University	390.86	34.17
Salinas	377.25	21.44
WHU-Hi-LongKou	392.87	35.21
WHU-Hi-HanChuan	412.23	54.35

E. Ablation Experiment

An ablation analysis is conducted to quantify the effects of the FMSEM, GLFFM, and DWCAM on classification performance. By successively introducing each module and observing the corresponding changes in OA and AA, the study clarifies the role and significance of each component in improving the model's performance. The quantitative experimental results can be found in Table IX.

First, the effectiveness of individual modules is examined. In case 1, only FMSEM is employed, achieving OAs of 94.52%, 95.23%, 94.82%, and 94.61%, with corresponding AAs of 92.14%, 94.85%, 93.54%, and 93.22%. This demonstrates that FMSEM effectively captures discriminative spectral signatures, confirming the importance of spectral enhancement; however, the absence of spatial modeling limits performance in spatially heterogeneous regions. Case 2 uses only GLFFM, yielding slightly lower OAs (94.24%–95.05%) but emphasizing local spatial consistency. Case 3 relies solely on DWCAM, with OAs dropping to 92.85%–93.54%, indicating that global semantic aggregation alone is insufficient without refined spectral–spatial features. The single-module experiments clearly demonstrate that while each module has certain effects, none can fully realize its potential when used independently.

Building upon this foundation, the dual-module combinations show significant performance improvements, confirming the complementary nature of the modules. In case 4, only GLFFM and DWCAM are employed, without FMSEM. The OAs on Pavia University, Salinas, WHU-Hi-LongKou, and WHU-Hi-HanChuan are 96.89%, 97.63%, 97.53%, and 97.49%, respectively, with corresponding AAs of 95.81%, 97.76%, 97.49%, and 97.33%. These results indicate that while GLFFM stabilizes spatial representations and DWCAM captures high-level semantic dependencies, the absence of FMSEM limits the network's ability to enhance critical spectral structures and capture multiscale frequency information.

In case 5, FMSEM is introduced while GLFFM is removed and DWCAM is retained. OA improves to 98.17%, 98.86%,

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TABLE IX
ABLATION ANALYSIS RESULTS ON PAVIA UNIVERSITY, SALINAS, WHU-HI-LONGKOU, AND WHU-HI-HANCHUAN

Cases	Components			Indicators							
				Pavia University		Salinas		WHU-Hi-LongKou		WHU-Hi-HanChuan	
	FMSEM	GLFFM	DWCAM	OA (%)	AA (%)	OA (%)	AA (%)	OA (%)	AA (%)	OA (%)	AA (%)
1	✓	×	×	94.52	92.14	95.23	94.85	94.82	93.54	94.61	93.22
2	×	✓	×	94.24	91.83	95.05	94.12	94.63	93.25	94.32	92.94
3	×	×	✓	92.85	89.56	93.54	91.23	93.12	90.84	92.93	90.52
4	×	✓	✓	96.89	95.81	97.63	97.76	97.53	97.49	97.49	97.33
5	✓	×	✓	98.17	97.68	98.86	98.68	98.74	98.51	98.76	98.54
6	✓	✓	×	98.31	98.79	99.24	99.12	99.07	98.92	99.15	99.02
7	✓	✓	✓	**99.89**	**99.84**	**99.96**	**99.93**	**99.91**	**99.76**	**99.89**	**99.79**

98.74%, and 98.76%, and AA to 97.68%, 98.68%, 98.51%, and 98.54%, demonstrating that FMSEM effectively strengthens spectral discriminability, providing more reliable inputs for semantic aggregation via DWCAM. The absence of GLFFM, however, still constrains local spatial feature refinement.

Similarly, in case 6, FMSEM is combined with GLFFM, while DWCAM is removed. The results show that OA reaches 98.31%, 99.24%, 99.07%, and 99.15%, and AA reaches 98.79%, 99.12%, 98.92%, and 99.02%. These results indicate that the spectral enhancement provided by FMSEM, together with the global–local spatial fusion from GLFFM, collaboratively improves multiscale feature representations, enhancing fine-grained spectral–spatial feature extraction. However, the absence of DWCAM limits the model's capacity for high-level global semantic aggregation.

Finally, case 7 employs all three modules. OA and AA exceed 99% across all datasets, reaching 99.89%, 99.96%, 99.91%, and 99.89% (OA) and 99.84%, 99.93%, 99.76%, and 99.79% (AA). Compared to case 4 (without FMSEM), substantial OA improvements are consistently observed across all datasets, with increases of 3.00% on Pavia University, 2.33% on Salinas, 2.38% on WHU-Hi-LongKou, and 2.40% on WHU-Hi-HanChuan. These results clearly demonstrate that FMSEM is not merely an incremental module, but a foundational component of the proposed framework. This significant performance improvement highlights the critical role of frequency-domain spectral enhancement in extracting highly discriminative features. This confirms that FMSEM, GLFFM, and DWCAM work complementarily to jointly optimize spectral enhancement, spatial stabilization, and semantic aggregation. The full integration of multiscale, multifrequency, and local–global information enables the model to achieve the highest classification performance, validating FMSEM as a core module and demonstrating the cooperative effectiveness of all three components.

F. Convergence and Stability Analysis of the Kernel Refinement Strategy in GLFFM

To validate the effectiveness of the proposed kernel refinement strategy in GLFFM, additional experiments were conducted on the Pavia University dataset to compare the proposed strategy with a fully learnable dynamic convolution design from the perspectives of convergence behavior and optimization stability. Specifically, while keeping the overall network architecture unchanged, only the kernel refinement strategy adopted in GLFFM was replaced with a fully learnable dynamic convolution design for comparison. All other components, training settings, and hyperparameters were kept identical to ensure a fair evaluation. To analyze the convergence behavior of the two designs, the training loss curves were recorded during the optimization process, as shown in Fig. 17. As observed, the proposed kernel refinement strategy exhibits more favorable convergence behavior throughout the training process compared with the fully learnable dynamic convolution design. From the perspective of convergence speed, the proposed kernel refinement strategy consistently maintains lower training loss over training stages and exhibits a faster decreasing trend. This indicates that the proposed strategy facilitates more efficient parameter optimization and accelerates the model convergence process. In terms of optimization stability, the proposed kernel refinement strategy exhibits a smoother optimization trajectory during training. Compared with the fully learnable dynamic convolution design, its training loss curve shows smaller fluctuations, indicating a more stable optimization process. This behavior can be attributed to the introduction of a frozen convolution kernel as a stable structural prior in the proposed kernel refinement strategy, which reduces the effective optimization

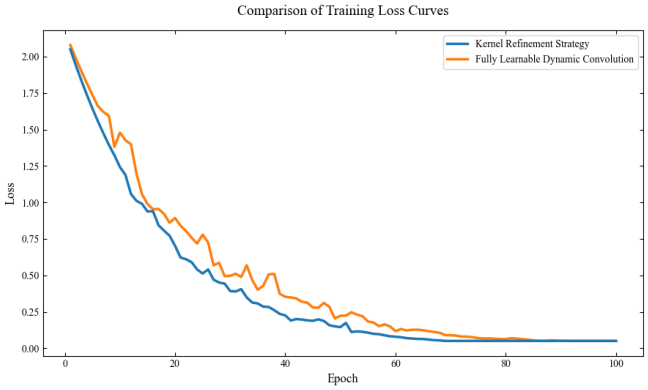


Fig. 17. Comparison of training loss convergence curves between the proposed kernel refinement strategy and a fully learnable dynamic convolution design on the Pavia University dataset.

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space and constrains kernel learning within a stable reference framework, thereby facilitating more efficient and robust parameter optimization.

V. Conclusion

In this work, we proposed FMSE-DGAF, a novel HSI classification framework that progressively extracts and refines spectral–spatial features through frequency-domain spectral enhancement and dynamic global–local attention. By jointly incorporating the FMSEM, the GLFFM, and the DWCAM, the proposed framework enables complementary modeling of spectral discrimination, spatial stability, and high-level semantic dependencies. Results obtained from five benchmark datasets with a wide spatial resolution range (0.109–30 m, approximately a 300-fold variation) confirm that FMSE-DGAF consistently performs well in classification tasks, demonstrating strong generalization capability. Furthermore, ablation studies confirm that FMSEM serves as the foundational component of the proposed framework, while GLFFM and DWCAM provide complementary contributions to overall feature classification performance. In addition, compared with recent Mamba-based models, the proposed FMSE-DGAF achieves a favorable balance between classification accuracy and computational efficiency, making it a promising high-efficiency alternative to Mamba-based models for HSI classification. Moreover, the proposed FMSE-DGAF exhibits low inference memory overhead, indicating its promising deployment potential in resource-constrained environments and practical HSI classification tasks. Despite the promising performance achieved by FMSE-DGAF, certain limitations remain for further exploration. FMSE-DGAF introduces additional computational overhead due to frequency-domain operations and dynamic attention mechanisms. To further improve computational efficiency and scalability, future extensions of this work may incorporate structured pruning for both convolutional and attention-based components, as well as adaptive or sparse attention mechanisms. In particular, the learned attention representations in DWCAM (e.g., the attention matrix $\mathbf{A}$ and the channel-wise scaling factor $\mathbf{S}$) could potentially serve as importance indicators to guide structured pruning. This could facilitate a more efficient deployment of FMSE-DGAF in large-scale and real-world HSI classification scenarios.

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